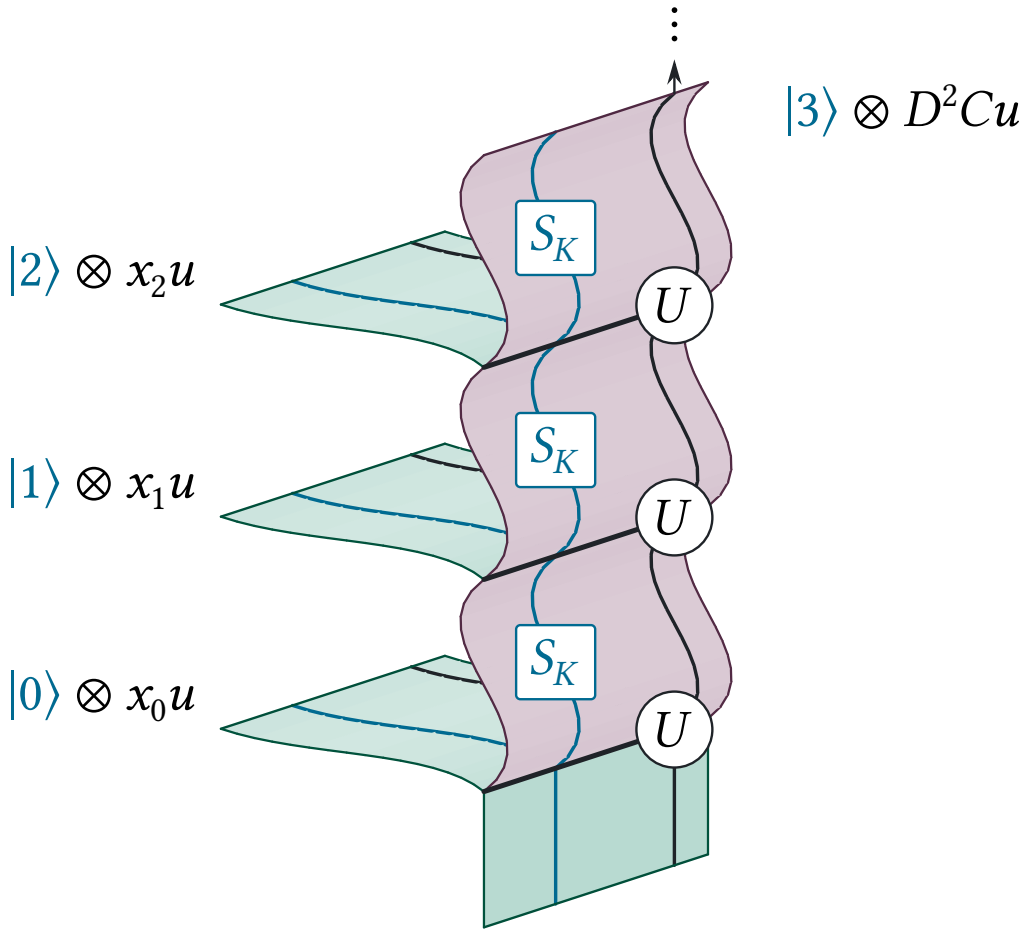
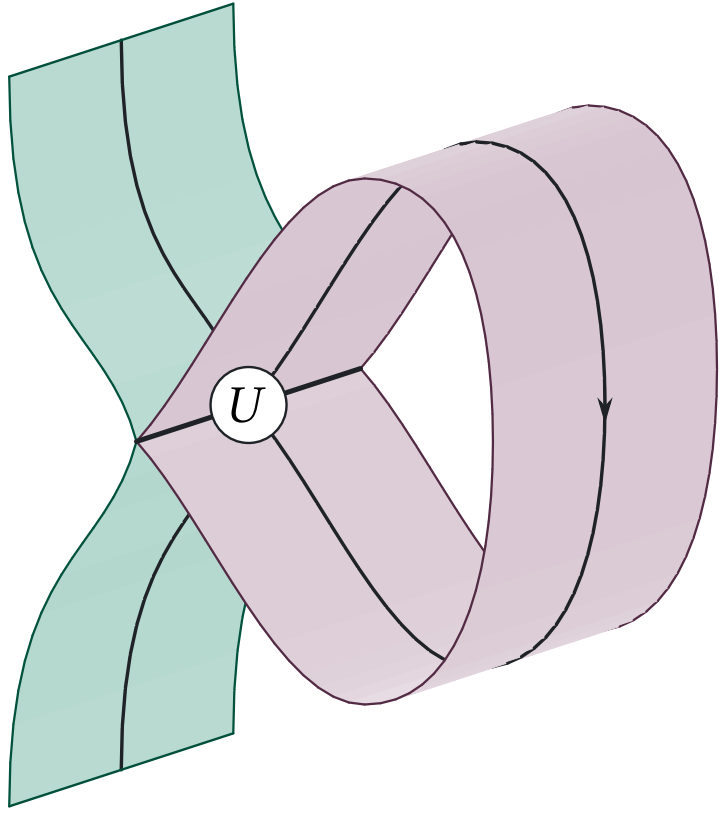
(a) Keep the count



$\mathrm{alloc}_0(u) = |0\rangle \otimes u$

(b) Coherent feedback

$Lu = (x_0 + x_1 + x_2 + \cdots)u$



u